

An
Industry Oriented Mini Project Report
On

Face Recognition Attendance System

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project report titled **“FACE RECOGNITION ATTENDANCE SYSTEM”** is being submitted by **K. Nithin (227Z1A0576), L.Bindu (227Z1A0591) and K. Praveen (227Z1A0582)** in Partial fulfillment for the award of **Bachelor of technology in Computer Science & Engineering** is a record bonafide work carried out by them. The results embodied in this report have not been submitted to any other University for the award of any degree.

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DECLARATION

We K. Nithin, L. Bindu and K. Praveen the students of **Bachelor of Technology in Computer Science and Engineering, Nalla Narasimha Reddy Education Society's Group of Institutions**, Hyderabad, Telangana, hereby declare that the work presented in this project work entitled **FACE RECOGNITION ATTENDANCE SYSTEM** is the outcome of our own bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of engineering ethics. It contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning.

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ABSTRACT

The "Face Recognition Attendance System using Python" provides a contactless and automated solution for attendance management in workplaces, schools, and events. Developed using Python, the system incorporates OpenCV for real-time image processing, Face Recognition for precise facial detection, and NumPy for computational tasks. It features real-time face detection and recognition through camera feeds using pre-trained models, automatic attendance logging with timestamps stored in secure databases, and an easy registration process for capturing facial data and personal details. Additional functionalities include real-time notifications for attendance confirmation or alerts for unregistered users, detailed attendance reports and trend analytics through a user-friendly interface, and scalability to accommodate large populations while integrating with management systems like payroll or LMS. By addressing challenges such as impersonation, buddy punching, and manual errors, this system enhances efficiency and accuracy. Future developments may include multi-camera support, mobile integration, and advanced features such as emotion detection and mask recognition, making it a cost-effective and versatile solution for various applications.

Keywords:

Image processing, Face detection, Face recognition, Automated attendance, Secured database.

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